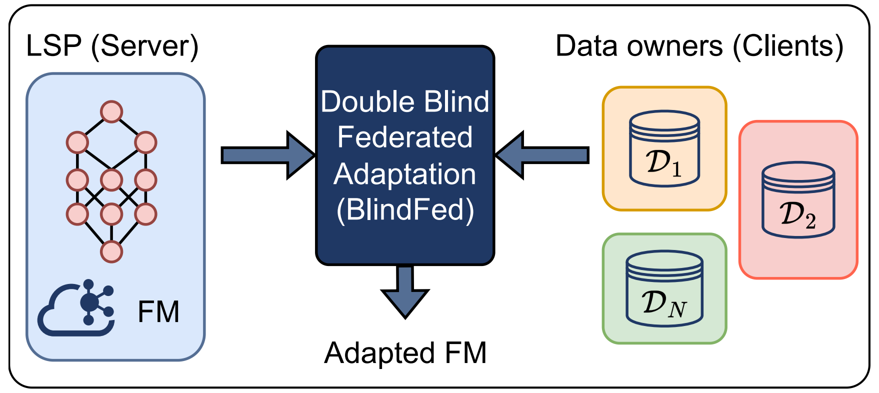
A Framework for Double-Blind Federated Adaptation of Foundation Models

Background



Double-Blind Constraints

Model Privacy: Clients cannot access the FM.
Data Privacy: The LSP cannot access the local data.

Goal

Train a lightweight **parallel adapter** and a **classification head** to maximize performance while preserving privacy.

Method

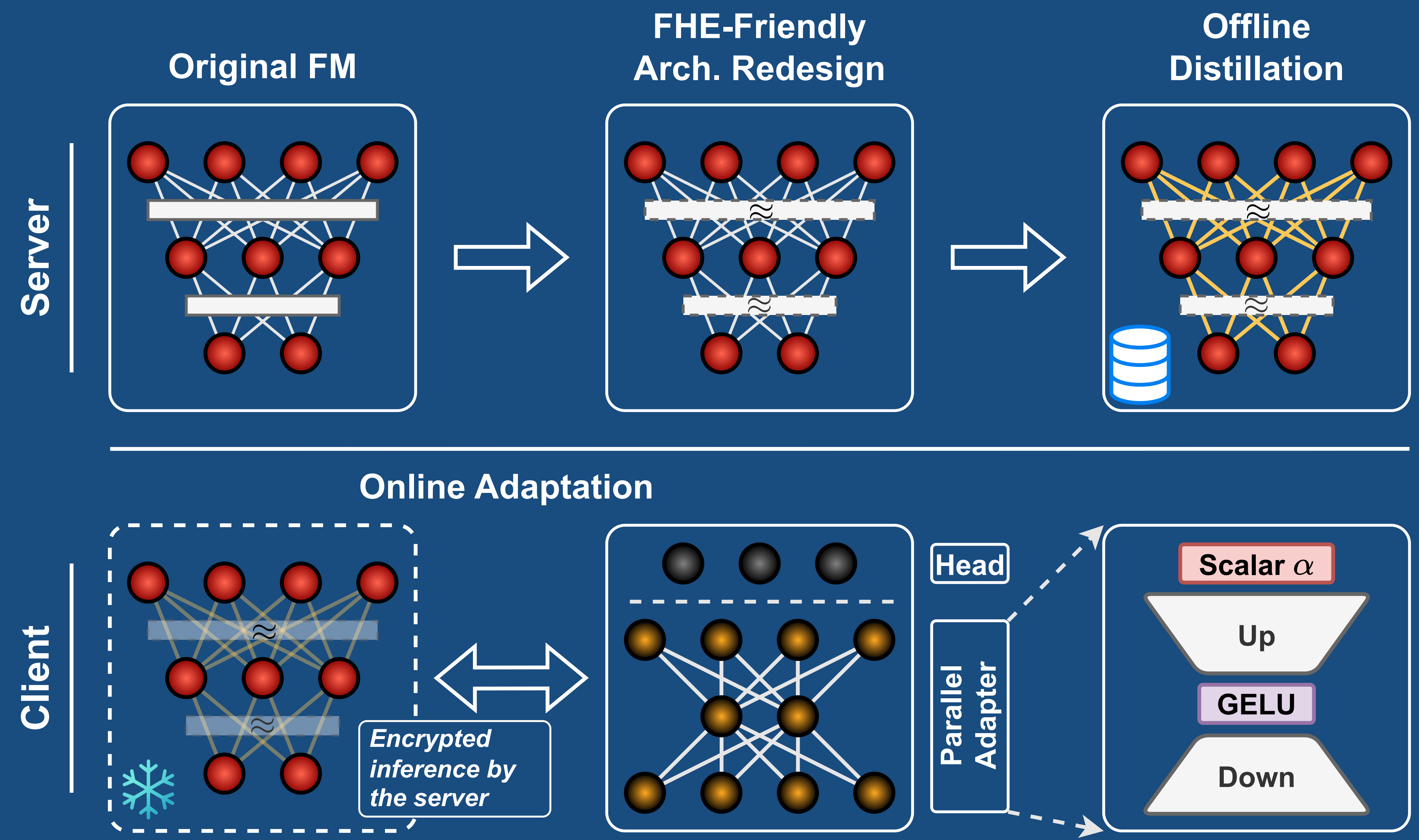
BlindFed: Recipe Card

Assumptions. Public auxiliary $\mathcal{D}_{\text{aux}}$; modular FM with L blocks; thin clients, powerful server; semi-honest parties.

Protocol.

1. **Pre-processing:** Server: Convert FM to FHE-friendly form (approximate Softmax, GELU, LayerNorm) and distill from original FM on $\mathcal{D}_{\text{aux}}$. Clients: Send FHE encrypted data to the server, only once.
2. **Per-round:**
   (a) Server performs encrypted block-wise inference and permutes IR (intermediate repr.) batches.
   (b) Clients decrypt IRs and train parallel adapter + head locally (no FM exposure).
   (c) Server aggregates updates via MPC; optional SBS.

Double-Blind Adaptation of Foundation Models



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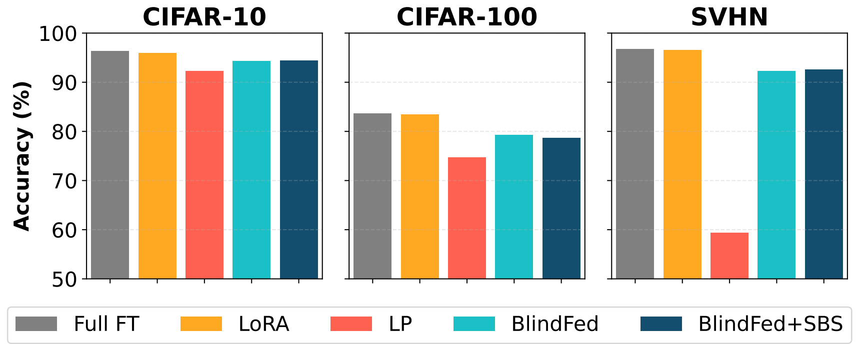
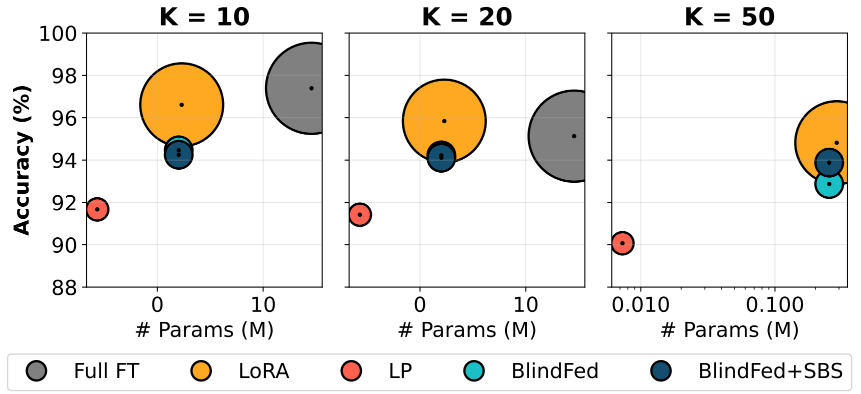
Experiments

Main Results



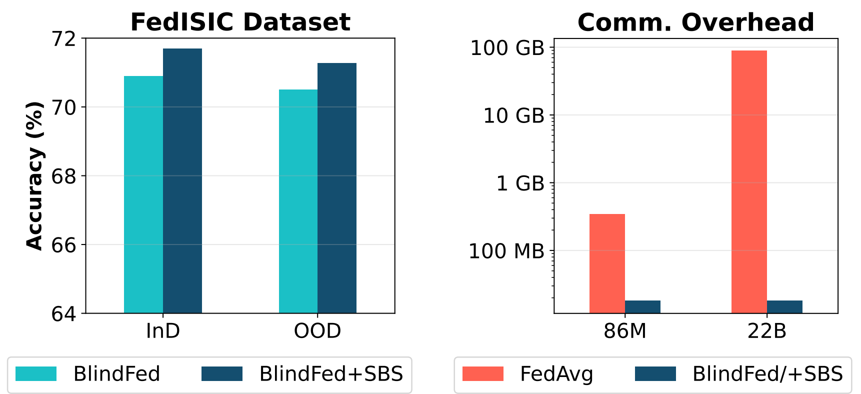
BlindFed significantly outperforms DB-LP, while achieving accuracy only marginally lower than non-DB methods.

Scaling with Number of Clients



BlindFed achieves strong performance across different no. of clients with far lower latency and parameter count.

OOD Auxiliary Data & Overhead



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